

Literature Answer to the $C[0, 1]$ Pre-Komlos Question in arXiv:1509.07914

fa_banach_001, agent_lane_17

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Status

This is a lightweight `literature_already_answered` packet. It records that the question whether $C[0, 1]$ has the pre-Komlos property, asked by Gao, Troitsky, and Xanthos in *Uo-convergence and its applications to Cesaro means in Banach lattices*, arXiv:1509.07914, was answered by a separate later paper.

Source Question

After Example 5.3, the source paper states:

“We do not know whether $C[0, 1]$ has the pre-Komlos property, or more generally, when $C(K)$ has the pre-Komlos property.”

The source definition uses norm-bounded sequences and requires a subsequence whose every further subsequence has Cesaro means that are uo-Cauchy.

Supporting Answer

The supporting paper is E. Y. Emelyanov, N. Erkursun-Ozcan, and S. G. Gorokhova, *Komlos properties in Banach lattices*, arXiv:1710.02580.

In Section 2, the supporting paper says that the *nuo*-Komlos and *nuo*-pre-Komlos properties were introduced in arXiv:1509.07914 under the names Komlos and pre-Komlos. In Section 3 it explicitly notes that arXiv:1509.07914 left the $C[0, 1]$ case unknown.

The decisive result is Section 3, Theorem 3, followed by Corollary 2. It states that if K is a compact metric space possessing a nonempty separable open subset without isolated points, then $C(K)$ fails the relevant pre-Komlos property. Since $K = [0, 1]$ is compact metric and has no isolated points, the source question has a negative answer:

$C[0, 1]$ does not have the pre-Komlos property.

Scope

This packet records an already-known literature answer, not a new result from the run. It answers the explicit $C[0, 1]$ question and a large class of compact metric $C(K)$ examples. It does not classify all compact Hausdorff spaces K for which $C(K)$ has the pre-Komlos property.

Search Bounds

The literature check used the exact phrases

`pre-Komlos property C[0,1]`, `pre-Komlos C(K)`, `pre-Komlos uo-Cauchy`,

and author/title searches around Gao–Troitsky–Xanthos and “Komlos properties in Banach lattices”. These searches found arXiv:1710.02580 as the direct later answer.

References

1. N. Gao, V. G. Troitsky, and F. Xanthos, *Uo-convergence and its applications to Cesaro means in Banach lattices*, arXiv:1509.07914, 2015.
2. E. Y. Emelyanov, N. Erkursun-Ozcan, and S. G. Gorokhova, *Komlos properties in Banach lattices*, arXiv:1710.02580, 2017.